

Core Java - Fundamentals

- Variables: Local, Instance, Static
 - Data Types: Primitive (int, double, etc.) & Non-Primitive (String, Array)
 - Operators: Arithmetic, Relational, Logical, Ternary
 - Control Statements: if-else, switch
 - Loops: for, while, do-while, break, continue
 - Arrays: Single & Multi-dimensional, length, traversal
 - Strings: Immutable, equals() vs ==, common methods (length, substring, split)
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OOP Concepts

- Class & Object: Blueprint & Instance
 - Constructor: Same name as class, used for initialization
 - Encapsulation: Private variables + Getters/Setters
 - Polymorphism: Overloading (compile-time), Overriding (runtime)
 - Inheritance: extends keyword, single/multilevel/hierarchical
 - Abstraction: Interface & Abstract class
 - this: Refers to current object
 - static: Class-level member shared by all objects
 - final: Constant variable, no override, no inheritance
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Exception Handling

- Exception: Runtime error disrupting normal flow
 - Checked vs Unchecked Exceptions
 - Common Runtime Exceptions: Arithmetic, NullPointerException, ArrayIndexOutOfBoundsException
 - try-catch-finally block
 - throw vs throws
 - Custom Exception basics
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Wrapper Classes

- Primitive ↔ Object conversion (Autoboxing/Unboxing)
- String → int: Integer.parseInt()
- String → double: Double.parseDouble()
- Primitive → String: String.valueOf()
- Common Exception: NumberFormatException

Type Casting

- Primitive Casting: Widening (automatic), Narrowing (manual)
 - Object Casting: Upcasting (Child → Parent), Downcasting (Parent → Child)
 - instanceof for safe downcasting
 - ClassCastException risk
 - Reference decides access, Object decides behavior
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Collections Framework

- ArrayList: Ordered, Duplicates allowed, Index supported
 - HashSet: No duplicates, No order, No indexing
 - HashMap: Key-Value pairs, No duplicate keys
 - Iterator usage
 - Generics for type safety
 - Collections store Objects only (use Wrapper classes)
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